



Volatility forecast of country ETF: The sequential information arrival hypothesis

Tseng-Chan Tseng^a, Chien-Chiang Lee^b, Mei-Ping Chen^{c,*}

^a Department of Finance, Ningbo Institute of Technology, Zhejiang University, Ningbo, China

^b Department of Finance, National Sun Yat-sen University, Kaohsiung, Taiwan

^c Department of Accounting Information, National Taichung University of Science and Technology, Taichung, Taiwan

ARTICLE INFO

Article history:

Accepted 21 February 2015

Available online 13 March 2015

Keywords:

Exchange-traded funds

Volatility forecast

Sequential information arrival hypothesis

Heterogeneous autoregressive model

Realized range-based volatility

Realized range-based bi-power variance

ABSTRACT

This paper uses daily data to examine whether the sequential information arrival hypothesis is supported in single country Exchange Traded Fund (ETF) market, and to model the forecast of ETF's volatility. The work is based on incorporating lagged trading volume into the 'heterogeneous auto-regressive' (HAR) model of regression realized range-based volatility (RRV) on realized range-based bi-power variance (RBV) (HAR–RRV–RBV–cum–Vol model, hereafter), in an attempt to improve the overall forecast of realized variance. We find that the forecasting performance of the HAR–RRV–RBV–cum–Vol model is better than other models for both in-sample and out-of-sample forecasts. The results support the sequential information arrival hypothesis in single country ETF market, by which lagged volume is available to predict current volatility.

© 2015 Elsevier B.V. All rights reserved.

1. Introduction

As a useful vehicle to gain benefits from international diversification or hedge against foreign market risk in the long run, the US Exchange Traded Funds (ETFs) account for around US\$1 trillion, or 70% of global ETF assets, equal to around one-quarter of aggregate turnover in US equities (Kosev and Williams, 2011).¹ In particular, single country ETF, a sub-sector of the ETF market, is designed to track stock market indices of foreign countries. A special feature of country ETFs is that ETF shares and their underlying portfolios are traded separately in two different markets: The ETF is traded in the US when the underlying portfolio is traded in the foreign home-country. Due to the strong development and importance of ETFs, scholars research many relative ETF issues, such as price discovery and forecast (Krause and Tse, 2013; Levy and Lieberman, 2013), efficiency and performance (Deville, 2008), competition or substitution effect of ETFs and their related index funds (Barnhart and Rosenstein, 2010; Rompotis, 2009), Tracking errors (Shin and Soydemir, 2010), and behavior of volatility (Aber et al., 2009; Chelley-Steeley and Park, 2011; Rompotis, 2010).

Since the raised significant influence of volatility risk in modern pricing and risk management theories (Çelik and Ergin, 2014; McAleer and Medeiros, 2008), more and more extensive studies attempt to explore

the volatility of asset pricing, diversify investment, and risk management (Byun and Kim, 2013). Many scholars have focused on evaluating the performances among various volatility forecast models. For example, Andersen and Bollerslev (1998); Barndorff-Nielsen and Shephard (2006), and Liu and Wan (2012) show that a daily realized volatility (RV), as calculated by the sum of the squared returns over specified intervals within a sampling period, is less subject to measuring error and thus less noisy. Lately, Christensen and Podolskij (2006) propose the realized range-based variance (RRV), which replaces every squared return of RV, and use the estimation of the quadratic variation (QV) of a continuous sample path semi-martingale, is five times more accurate than traditional measures of RV.² Moreover, Akay et al. (2010) suggest that range-based volatility measures not only reduce the disturbance of microstructure noise, but are also better to differentiate liquidity pressure from regular trading.

There exists a great deal of studies which utilize RRV to forecast, for example, oil price volatility (Tseng et al., 2009), exchange markets (Wang and Yang, 2009), S&P depositary receipts (Liu et al., 2012), and the S&P 500 index (Chung et al., 2011). However, the RRV, modeling and forecasting the volatility of ETF prices, is relatively less concerned. We use country ETF data for the following reasons: ETFs are usually more liquid and thereby more suited to our research question as

* Corresponding author. Tel.: +886 4 22196020; fax: +886 4 22196131.

E-mail address: meiping@nutc.edu.tw (M.-P. Chen).

¹ ETFs are popular investment vehicles because they offer investors continuous trading during exchange hours (high liquidity), low premiums or discounts, tax efficiency, diversification benefits, low annual fees compared to the fees of mutual funds, and transparency (Miffre, 2007; Small et al., 2012).

² Li et al. (2010) suggest that RRV presents a relatively stable property on long-term memory and is less influenced by sampling frequency. Martens and van Dijk (2007) indicate that realized range estimator will be biased depending on the sampling frequency and due to infrequent trading.

volatility can lead to financial intermediaries liquidating more liquid assets from their portfolios (Brunnermeier, 2009). ETFs can be simply purchased and sold by all investors at lower transaction cost (Marshall et al., 2013). Tse and Martinez (2007) and Ackert and Tian (2008) show that the premium of country ETFs is larger in magnitude and more persistent, compared with other ETF sectors.

Thus, it is particularly important and appealing to academics, investors, and practitioners to investigate the performances of different volatility forecast models for single country ETFs, which is relatively little work undertaken. Except Liu et al. (2012) employ the Generalized Autoregressive Conditional Heteroskedasticity (GARCH) models, to forecast volatility for Standard & Poor's Depositary Receipts.

This paper contributes to the existing related studies with three significant respects. First, the majority related studies measure volatility either within a country or across-countries. International evidence is an integral part of the empirical test of economic theories (Wang and Yang, 2013). However, the possible effects of non-synchronized trading hours, divert country policy setting, foreign exchange, and trading constraints should take into account when study cross-country issue. Additionally, global investors, economic policymakers, and market regulators tend to focus on cross-country market-wide indicators, such as volatility.

Moreover, US single country ETFs are traded in the US market and are designed to track a wide variety of foreign country indices (Levy and Lieberman, 2013); thus country ETFs make it easy for investors to achieve global diversification without directly investing in foreign stock markets for benefits. The exchange risk can be avoided, and the barriers of directly investing in foreign stock markets can be eliminated (Meric et al., 2008). As such, country ETFs can be useful to reduce exposure to foreign exchange market risk in the long-term and for a relatively extended investment horizon (Levy and Lieberman, 2013). Due to significantly lower expense ratios, iShares country ETFs offer higher returns than corresponding closed-end country funds and have higher Sharpe ratios (Deville, 2008). Thus, this study considers volatility of single country ETFs aiming at the cross-country level. We broaden the study of volatility forecast to furnish better understanding in terms of the volatility of single country ETFs by taking a global perspective.

Second, a majority of previous studies generally tend to use intraday data. However, high frequency data may not be capable of depicting the long-term interest of investors. In practice, the presence of so-called market microstructure features, raised especially for high-frequency data, generates complications (Chaboud et al., 2010). Compared with Mizrahi (2008), who applies daily data to investigate home equity derivatives, we take the sampling interval with weekly changes so as to show that these methods, even at a weekly frequency, are accurate in predicting ETF market movements. The daily range-based volatility estimators, which are easily derived from readily available opening, high, low and closing prices, also offer a useful alternative (Garman and Klass, 1980; Liu et al., 2012). Ghysels et al. (2005) suggest that exploring data with various frequencies can sharpen the estimates of conditional volatility. Afterwards, we provide a more efficient model, using lags 1, 5, and 20 weeks respectively, in forecasting 1- and 5-week volatilities of ETFs, which provides an additional insight to the volatility forecast with lower frequency.³

Last but not the least, as opposed to using return-based volatility (RV) as a proxy for true volatility and a GARCH model on volatility

forecast, the study regards RRV as a proxy for true volatility and proposes to apply the heterogeneous auto-regressive model (HAR) developed by Corsi (2004).⁴ The consistency of RV breaks down when market microstructure noise contaminates high-frequency asset prices (Hansen and Lunde, 2006). RRV presents a relatively stable property on long-term memory and is less influenced by sampling frequency (Li et al., 2010). HAR is able to reproduce the memory persistence observed in the data and to reduce the number of parameters to estimate; and their prediction ability performs better than autoregressive fractionally integrated moving average.

Additionally, Stock and Watson (2004) find that a combination of model-based forecasts is a dominant approach and often out-performs the vest individual forecasts. It has been found in several recent studies that the use of daily realized range-based bi-power variance (PBV) as the regressor provides good performance in the prediction of future volatility, since it is immune to jump (Andersen et al., 2007; Ghysels et al., 2005).

Todorova and Souček (2014) evidence the incremental value of volume. Therefore, we incorporate lagged trading volume into the HAR–RRV–RBV model (i.e., HAR–RRV–RBV-cum-Vol model) to facilitate our test of 21 ETFs so as to test whether the Sequential information arrival (SIA) hypothesis is supported. Particularly, the SIA hypothesis is based on frequent observations of gradual information release in the market (Zhang, 2007). We adopt the SIA hypothesis for improving testing the effect of trading volume on the persistence of volatility of returns. We refer to this new forecasting model as the HAR–RRV–RBV-cum-Vol model. That is, we incorporate lagged trading volume into the HAR model of range-based bi-power variance regressor on realized range-based volatility on both in-sample and out-of-sample and both cases of with and without trading volume to explore the information content of lagged volume in explaining ETF volatility.⁵ Our particular approach can not only be intuitively appealing to determine the impact of lagged volume existed, but also possibly prove to yield a beneficial improvement in terms of volatility forecast itself.

This paper complements the literature on both of using model-based forecasts to ETF volatility with 1, 5, and 25 weekly high-low statistics and the modeling of the impact of volumes while using the HAR–RRV model. We find strong evidence that the HAR–RRV–RBV-cum-Vol models are more efficient in capturing the long-term volatility of single country ETFs than others. Moreover the jump components do not contribute to future valuable information. Finally, all of the above results hold for in-of-sample and out-of-sample. The out-of-sample forecasting precision, which is perhaps the ultimate measure of a model's forecasting potential, implies that our results are unlikely to be due to sampling error or over-fitting. That is, the inclusion of ETF trading volume in HAR–RRV–RBV models improves the forecast accuracy, while modeling trading volume is considered directly into our formulations. In particular, trading volume has a significant effect on future volatility. The results support the sequential information arrival hypothesis in single country ETFs, implying that high trading volume causes high return volatility.

The remainder of the paper proceeds as follows. Data and the sampling process are described in Section 2. Section 3 discusses the methodology. Section 4 presents the empirical findings. Section 5 offers the concluding remarks.

2. Data

We collect the daily open, close, high, and low prices, as well as the daily trading volume (Vol, expressed in thousands of shares) for

³ There are two reasons for using lags 1, 5, and 20 weeks, respectively: First, Corsi (2004) demonstrates that the HAR–RV model is more capable of capturing the long-term memory property of realized variance. The HAR–RV model proposed by Corsi (2004) provides an additive cascade of three volatility components generated by the actions of different types of market participants. The three primary volatility components include short-term (daily), medium-term (weekly), and long-term (monthly) volatilities. Secondly, one of the specialties of ETFs is low turnover, implying that investors hold ETFs for a long period. Based on the weekly frequency used in this study, we thereby follow the setting for the volatility expectations of the HAR–RV model proposed by Corsi (2004) that uses lags one, five, and twenty weeks, respectively, to forecast 1- and 5-week volatilities of ETFs.

⁴ HAR is thought of some past variations and of regressors to predict future increments in quadratic variations.

⁵ Two hypotheses examine the relationship between trading volumes and return volatility of equities. That is, the mixture of distribution hypothesis of Clark (1973) that volume and volatility are related contemporaneously, and the SIA of Copeland (1976) and Smirlock and Starks (1988) that volume and volatility possess a lead-lag relation.

individual ETFs from Datastream. The samples of single country ETFs are obtained from www.eft.com. Table A1 of Appendix A reports the profiles – including name of ETFs, ticker, and the number of observations. Our data include 21 ETFs across 19 countries for the period from 2000/01/01 to 2013/04/30, with the exception of 15 ETFs shorter periods used in each of the analyses due to the leakage of data (see Table A1 of Appendix A). The 19 ETFs are from Australia, Brazil, Belgium, Canada, China, France, Germany, Hong Kong, Italy, Japan, Malaysia, Mexico, Netherlands, South Korea, Spain, Sweden, Switzerland, Taiwan, and the UK. Herein the period from 2012/01/06 to 2013/05/24 is specified as out-of-sample evaluation, while the model parameters are estimated from 2000/01/07 to 2011/12/30. The time frame provides an interesting mix of bull and bear market trends, therefore incorporating an environment containing diverse market dynamics. Our data processing procedure basically follows the approach in Andersen et al. (2001).

Alizadeh et al. (2002) point out that the microstructure noise of markets has adverse effect on the high frequency estimator. The data with higher frequency may be contaminated with market microstructure noise.⁶ Thus, we use daily data, which allow us to draw more inferences. Vol indicates the number of shares traded during the day, and is the sample standard deviation of the daily returns over 1, 5, and 20 weeks.

The means of the realized volatility levels, including RRV_t , RBV_t , as well as the logarithm of trading volume, $\log(Vol_t)$, are presented in Table 1, in which the mean values of the RRV are much larger than the RBV. The mean of the weekly RRV ranges between 4.359 and 17.818; while that of the RBV ranges between 3.895 and 13.79. The iShare Xinhua China 25 (FXI) exhibits the highest mean of the log (Vol) and the iShare MSCI Netherlands (EWN) having the lowest. EWZ (iShare MSCI Brazil Capped ETF) has the highest return variance, and EWJ (iShare MSCI Japan) have the lowest across sampled ETFs in both RRV and RBV.

3. Methodology

3.1. Realized variance

Following Barndorff-Nielsen and Shephard (2004, 2006) and Andersen et al. (2007), we consider a scalar log-price $p(t)$ evolving within a continuous time stochastic volatility jump-diffusion process. One of the most important aspects of semi-martingales is the QV process, which is defined as:

$$[p](t) = p \rightarrow \lim_{n \rightarrow \infty} \sum_{j=0}^{n-1} (p(s_{j+1}) - p(s_j))^2 \quad (1)$$

for any sequence of partitions $0 = s_0 < s_1 < \dots < s_n = t$ with $\max_{1 \leq j \leq n} |s_j - s_{j-1}| \rightarrow 0$ for $n \rightarrow \infty$. The theory of stochastic integration states that QV is a natural measure of sample path variability for the cumulative return process, $r(t) \equiv p(t+H) - p(t)$ expressed as:

$$QV_{t,t+H} = \int_t^{t+H} \sigma^2(s) ds + \sum_{t < s < t+H} \kappa^2(s), \quad (2)$$

which contains the integrated variance and squared jumps.

We utilize daily data on 21 single country ETFs to predict future weekly RRV (from Monday to Friday). Following Christensen and Podolskij (2006), our estimates are based on an equidistant grid,

⁶ Microstructure noise generally implies that realized volatility and realized range measures are inconsistent estimators for the integrated variance, with the impact becoming more pronounced as the sample frequency increases (Bannouh et al., 2013).

Table 1

Descriptive statistics of realized range-based volatility and volume for the 21 sample ETFs.

Symbol	Variable means		
	RRV_t (%)	RBV_t (%)	$\log(Vol_t)$
EWA	7.456	5.668	7.546
EWK	6.541	4.253	5.300
EWO	7.564	5.202	5.668
EWZ	17.818	13.790	9.214
EWG	8.463	6.2481	7.337
EWQ	6.013	4.443	6.097
EWJ	8.684	5.763	7.668
EWI	8.202	6.191	8.450
EWL	5.981	4.369	5.697
EWU	4.359	3.303	10.487
EWM	6.688	4.710	7.871
EWN	5.975	3.895	5.115
EWY	9.211	6.878	7.972
EWP	5.975	4.430	5.637
EWD	8.508	6.047	5.661
EWL	5.565	4.166	5.784
EWT	10.391	7.229	9.019
EWU	6.781	5.181	7.388
EWV	12.553	9.418	8.012
FXI	11.069	8.775	10.642
PGJ	13.252	9.541	6.346

Note: RRV_t denotes realized range-based variance; RBV_t denotes realized range-based bi-power variance; and Vol_t denotes trading volume. The full name of 21 ETFs is presented in Table A1 of Appendix A.

$t_j = j/f$, $j = 1, \dots, f$, where f is the sampling frequency. The weekly range in this study is written as:

$$S_{t,j} = \text{Max}(p_h - p_l), \quad \text{for } j = 1, 2, \dots, f \quad (3)$$

where p_h (p_l) is the highest (lowest) price during each day. Thus, we have a total of 5 observations of p in every one-week period.

As suggested by Garman and Klass (1980), Parkinson (1980), and Liu et al. (2012), the daily range-based volatility estimators which are easily derived from readily available prices also offer a useful alternative, we use $N = \text{daily}$, since this corresponds to the daily sampling frequency used in this study. Using the measures of realized range-based variance as those used by Christensen and Podolskij (2006), we define weekly realized range-based variance (RRV) and realized range-based bi-power variation (RBV) as follows:

$$RRV_t^b = \frac{1}{\lambda_2} \sum_{j=1}^f (s_{t,j})^2 \quad (4)$$

$$RBV_t = \frac{1}{\lambda_1 \lambda_1} \sum_{j=2}^f S_{t,j} S_{t,j-1}, \quad (5)$$

where $\lambda r = 4/\sqrt{\pi}(1-4/2^r)2^{r/2}\phi((r-1)/2)\theta(r-1)$, for $r \geq 1$; $\phi(x)$ denotes the Gamma function and $\theta(x)$ the Riemann zeta function. Christensen and Podolskij (2006) indicate that RRV is downward-biased, with the exception of the Brownian semi-martingales, or $f = 1$, as presented by the superscript b .

According to Christensen and Podolskij (2006), we propose the following modified realized range-based estimator which is consistent for the $[p]_t$ of the jump-diffusion semi-martingales:

$$RRV_t \equiv \lambda_2 RRV_t^b + (1-\lambda_2) RBV_t \xrightarrow{P} [p]_t. \quad (6)$$

We define the first-order range-based bi-power variation as RBV_t , proving an alternative method of drawing inferences on the continuous path component of QV_t by:

$$RBV_t \xrightarrow{P} \int_0^t \sigma^2(t) dt. \quad (7)$$

Thus, Christensen and Podolskij (2006) suggest that the RBV is the most persistent process essentially because it is robust against the jump component.

3.2. HAR–RRV models

The basic idea behind the HAR model is that agents with different time horizons perceive, react to, and cause, different types of volatility components. As noted by Corsi (2004), the heterogeneity of agents may arise for a multitude of reasons, such as differences in their endowments, the degree of information, their prior beliefs, institutional constraints, temporal horizons, geographical location, risk profile, and so on. The HAR model concentrates on the heterogeneity which originates from the differences in the time horizons.

Corsi (2004) effectively demonstrates that the HAR–RV model has better ability in capturing the long-term memory property of realized variance. While, Christensen and Podolskij (2006) and Akay et al. (2010) claim that RRV is more efficient than RV. We thereby consider the HAR–RRV model to model the future volatility of the ETF exchanged in the U.S.

Christensen and Podolskij (2006) find that RBV_t , the analogous estimation of the continuous component of QV, is up to 7.2 times more efficient than the bi-power variation (BPV) proposed by Barndorff-Nielsen and Shephard (2004).⁷ This study thus employs RBV as a regressor for the prediction of realized volatility by proposing the following the HAR–RRV–RBV model:

$$RRV_{t,t+H} = \beta_0 + \beta_{1W}RBV_{t-1,t} + \beta_{5W}RBV_{t-5,t} + \beta_{20W}RBV_{t-20,t} + \varepsilon_{t,t+H}, \quad (8)$$

where H denotes the prediction horizons. We focus on the prediction of future realized volatility using 1 week and 5-week horizons ($H = 1$ and 5), respectively. There are different types of investors in financial markets. The different types of investors focus on the risk of different periods. Thus, this paper provides short- and middle-term predictions of future realized volatility. The measures of multi-period realized volatility, as proposed by Corsi (2004) and Andersen et al. (2007), are as follows:

$$RRV_{t,t+H} = (RRV_{t,t+1} + RRV_{t+1,t+2} + \dots + RRV_{t+H-1,t+H})/H \quad (9)$$

3.3. Modified HAR–RRV models

The recent studies by Lee and Rui (2002) and Chan et al. (2004) suggest that trading volume is positively related to volatility. The SIA hypothesis also provides some theoretical underpinnings for the analysis of the link between volume and volatility. According to the SIA hypothesis, originally proposed by Copeland (1976) and Jennings et al. (1981), the new information received by traders arrives in a sequential, random fashion, under the assumption that there are intermediate equilibria en route to the final equilibrium (Darrat et al., 2003). The responses to information by different traders represent the component parts of a series of incomplete equilibria. Thus, the SIA hypothesis suggests that lagged volume may be available to predict current volatility, and vice versa. That is, the main implication of the SIA hypothesis is that the sequential reaction to information suggests that asset price volatility is potentially predictable through the information transmitted from trading volume.

Support for the SIA hypothesis was provided by Darrat et al. (2003), who found that in a large number of Dow Jones Industrial Average (DJIA) stocks, trading volume, and return volatility followed a clear lead-lag pattern. It had been suggested earlier, by both Lamoureux and Lastrapes (1990) and Andersen (1996), that trading volume can be

regarded as a proxy for measuring the unobservable information flowing into a market. Darrat et al. (2003) also found the presence of significant causality flowing from trading volume to return volatility in at least 12 stocks, thereby providing further support for the SIA hypothesis. Recently, Darrat et al. (2007) pinpointed significant bi-directional causality between volume and volatility of stocks that is in accordance with the SIA hypothesis. Rannou and Barneto (forthcoming) revealed that significant contemporaneous and lagged positive causality of over-the-counter derivatives volume on spot/futures volatility, confirming the SIA hypothesis.

From a market microstructure perspective, price movements are caused mainly by the arrival of new information and the processes which lead to the incorporation of such information into market prices (Andersen, 1996). The stock price moves from one temporary equilibrium to the next in response to the arrival of large amounts of different information during each trading day. Consequently, the arrival of news containing more information tends to generate more concentrated trading volume during the information assimilation phase.

Furthermore, as noted by Andersen (1996), trading volume can be regarded as a proxy for measuring the amount of unobservable information flowing into the market. Thus, the daily trading volume implies a new information assimilation phase, and therefore, new price movements. The fundamental premise of the SIA hypothesis is that the lagged trading volume has not been completely assimilated into the prior change in price; in such a case, we may expect that prior volume, which contains relevant information un-included in prior volatility, can be regarded as the prediction of future volatility.

Our primary objective is to achieve improvement in the overall forecasting of volatility. However, when forecasting volatility in the next period, trading volume in the next period is still unknown; thus, only prior volume can be used to improve the volatility forecasting. The SIA hypothesis therefore provides the basic idea for a modified model of improvement to volatility forecasting. Following Darrat et al. (2003), we attempt to investigate whether lagged trading volume is available to predict current volatility.⁸

In this study we propose a refined model which possibly improves the predictability of future realized volatility. Adopting the SIA hypothesis, we incorporate lagged trading volume into the HAR–RRV–RBV model, and then propose the following specification to obtain our so-called HAR–RRV–RBV–cum-Vol model:

$$RRV_{t,t+H} = \beta_0 + \beta_{1W}RBV_{t-1,t} + \beta_{5W}RBV_{t-5,t} + \beta_{20W}RBV_{t-20,t} + \beta_{V1W}Vol_{t-1,t} + \beta_{V5W}Vol_{t-5,t} + \beta_{V20W}Vol_{t-20,t} + \varepsilon_{t,t+H}, \quad (10)$$

where Vol defines the logarithm trading volume. The measures of lagged volume, including $Vol_{t-1,t}$, $Vol_{t-5,t}$, and $Vol_{t-20,t}$, are also calculated as well in Eq. (9).

3.4. Out-of-sample forecast comparison

In order to compare forecast accuracy among several models, we apply the mean square error (MSE) criterion. Hansen and Lunde (2006) and Patton (2006) have emphasized the importance of using a robust criterion, such as MSE to compare forecasts between models. Moreover, we apply the modified Diebold and Mariano (1995, modified DM) test of equal forecast accuracy. Harvey et al. (1997) suggest that the modified DM test constituted the best available approach to the evaluation of the significance level of the observed differences between two forecasting sets. The reported p -values are based on the T test. As using the same estimation window, we only need to consider the influences from different parameters.

⁷ Andersen et al. (2007) argue that the jump component of realized variance is of no help whatsoever in predicting future volatility.

⁸ Darrat et al. (2003) suggest that there is significant causality flowing from trading volume to return volatility in at least 12 stocks, thereby providing further support for the SIA hypothesis.

4. Empirical results

4.1. In-sample results

Table 2 reports the in-sample HAR–RRV–RBV model adjust R^2 and MSE estimates, using the 21 single country ETF index data.⁹ We focus on two different prediction horizons, 1- and 5-week periods, i.e., $RRV_{t,t+H}$, for $H = 1$ and 5, respectively. The results reveal that ETF EWL achieves the lowest MSE, ranging from 37.331 at the 1-week horizon to 24.764 at the 5-week horizon. It should be noted that with an increase in the prediction horizon, there is a corresponding reduction in MSE for all ETFs. As noted by Forsberg and Ghysels (2007), RV is easier to predict since it is calculated over longer horizons which thereby provide smoother series.

Table 3 reports that the comparative ratios of the MSE of HAR–RRV–RBV-cum-Vol regressions are always lower than those of the HAR–RRV–RBV regressions. We have demonstrated that the HAR–RRV–RBV-cum-Vol model has more accurate predictive ability than the HAR–RRV–RBV model, which implies that the lagged trading volume variables do indeed measure the flow of relevant information. The economic interpretation of the HAR–RRV–RBV-cum-Vol model is that the future volatility expectations can be effectively constituted by various components of lagged volatility and lagged volume, most of which contain relevant information which has not been completely assimilated into the prior change in price.

4.2. Out-of-sample results

In terms of predictive ability, much greater accuracy is demonstrated by the in-sample fit measures of the HAR–RRV–RBV-cum-Vol model as compared to those of the HAR–RRV–RBV model. Since the primary aim of this study is to achieve overall improvement in the forecast of future ETF volatility, it is therefore important for us to determine whether the superior performance of the HAR–RRV–RBV-cum-Vol model is maintained in the out-of-sample forecasts.

Table 4 presents the out-of-sample results of the HAR–RRV–RBV and HAR–RRV–RBV-cum-Vol regressions. The model parameters are estimated with the data from 2000/01/07 to 2011/12/30. The out-of-sample forecasts are from 2012/01/06 to 2013/05/24. For both the HAR–RRV–RBV and HAR–RRV–RBV-cum-Vol regressions, the MSE is always lower for the HAR–RRV–RBV-cum-Vol models, except for EWO (iShare MSCI Australia Capped), EWG (iShare MSCI Germany), EWP (iShares MSCI Spain), EWD (iShares MSCI Sweden), EWU (iShare MSCI UK), and EWW (iShare MSCI Mexico). The results presented in Table 4 clearly reveal that most of ETF, the MSE results are lower in the HAR–RRV–RBV-cum-Vol regressions than in the HAR–RRV–RBV regressions. Following Narayan et al. (2014) as well as Narayan and Sharma (2014), we apply out-of-sample forecasting evaluation tests. Our results in Table 4 are also emphasized in the out-of-sample R-squared (OSR), as in the HAR–RRV–RBV-cum-Vol regressions that have a bigger R-squared than in the HAR–RRV–RBV regressions, especially in the 5-week prediction horizons.

Following Harvey et al. (1997), we conduct modified DM test to assess forecasting accuracy and the performances of the out-of-sample forecasts. The HAR–RRV–RBV models are taken as the benchmark models for all horizons in the modified DM test. If the test statistic is positive (negative), then this indicates that the model under examination is more (less) accurate than the benchmark model. Table 5 shows the out-of-sample forecasting performance test results employing t-statistic of the modified DM test. The results indicate that the majority of the prediction horizons of ETFs are positively significant at the 10% level in both 1- week and 5-week forecast horizons, implying that the HAR–RRV–RBV-cum-Vol model saliently outperforms the HAR–RRV–RBV model in terms of forecasting accuracy in the ETFs sampling.

Table 2

In-sample results of the HAR–RRV–RBV model.

ETF	Horizons			
	1 week		5 weeks	
	Adjust R^2	MSE	Adjust R^2	MSE
EWA	0.483	134.705	0.471	90.758
EWK	0.236	229.734	0.331	90.974
EWO	0.409	190.455	0.399	123.183
EWZ	0.585	419.343	0.460	394.704
EWG	0.458	151.909	0.433	96.468
EWQ	0.544	62.920	0.490	50.492
EWV	0.112	572.302	0.232	158.064
EWI	0.508	74.302	0.469	56.538
EWJ	0.500	48.513	0.493	33.589
EWK	0.415	40.086	0.354	28.521
EWM	0.165	128.530	0.276	44.654
EWN	0.147	281.913	0.272	84.424
EWY	0.461	148.408	0.468	98.996
EWP	0.526	57.124	0.473	45.188
EWD	0.357	188.519	0.436	92.507
EWL	0.471	37.331	0.468	24.764
EWT	0.224	271.838	0.353	101.880
EWU	0.513	82.818	0.480	62.133
EWW	0.308	383.491	0.353	190.410
FXI	0.595	231.179	0.558	193.527
PGJ	0.312	1105.669	0.250	648.055

Note: The table presents the adjust R^2 and mean squared errors (MSE) for one-week and five weeks in-sample RRV predictions for the 21 sample ETFs. The full name of 21 ETFs is presented in Table A1 of Appendix A.

In sum, lagged volume does enhance the performance of volatility forecast. Our results do support the implication of the SIA hypothesis. The sequential reaction to information suggests that ETFs volatility is potentially predictable via the information transmitted from trading volume. We find that lagged volume has a significant influence in explaining volatility for both the in-sample and out-of-sample results.

5. Conclusions

This is the first paper to show a linkage between trading volume and ETF volatility. The implication of this paper is not only in terms of the

Table 3

In-sample results of the HAR–RRV–RBV-cum-Vol model.

ETF	Horizons			
	1 week		5 weeks	
	Adjust R^2	MSE	Adjust R^2	MSE
EWA	0.490	132.639	0.485	88.330
EWK	0.239	228.631	0.341	89.390
EWO	0.411	188.917	0.409	120.404
EWZ	0.588	415.354	0.464	391.250
EWG	0.460	148.832	0.439	94.710
EWQ	0.560	61.680	0.499	49.509
EWV	0.115	560.388	0.235	155.382
EWI	0.515	73.790	0.477	55.772
EWJ	0.503	47.957	0.502	32.845
EWK	0.420	39.264	0.358	27.966
EWM	0.166	127.814	0.281	44.152
EWN	0.149	275.888	0.274	82.688
EWY	0.468	146.122	0.481	96.556
EWP	0.530	56.792	0.484	44.062
EWD	0.362	184.568	0.442	90.733
EWL	0.475	36.685	0.470	24.387
EWT	0.226	270.074	0.371	98.443
EWU	0.515	82.326	0.495	60.261
EWW	0.311	376.665	0.359	186.998
FXI	0.598	224.030	0.562	187.348
PGJ	0.315	1069.261	0.254	629.620

Note: The table presents the adjust R^2 and mean squared errors (MSE) for one-week and five weeks in-sample RRV predictions for the 21 sample ETFs. The full name of 21 ETFs is presented in Table A1 of Appendix A.

⁹ The parameter estimation results for the 21 single country ETF index are not reported here to save space.

Table 4

Out-of-sample results for the 21 ETFs using the HAR–RRV–RBV and HAR–RRV–RBV–cum–Vol models.

ETF	HAR–RRV–RBV model		HAR–RRV–RBV–cum–Vol model	
	Horizons		Horizons	
	1 week	5 weeks	1 week	5 weeks
EWA	3.403 (0.490)	5.948 (0.473)	3.156 (0.490)	5.463 (0.474)
EWK	5.166 (0.264)	4.940 (0.344)	4.974 (0.264)	4.602 (0.346)
EWO	7.818 (0.420)	7.028 (0.410)	7.774 (0.420)	7.518 (0.411)
EWZ	18.145 (0.593)	38.685 (0.462)	16.676 (0.593)	35.312 (0.465)
EWQ	5.633 (0.475)	7.772 (0.440)	5.223 (0.475)	6.929 (0.442)
EWG	7.477 (0.551)	4.007 (0.491)	7.435 (0.551)	3.964 (0.493)
EWJ	20.138 (0.118)	20.398 (0.233)	18.104 (0.118)	24.671 (0.235)
EWI	4.384 (0.561)	6.788 (0.502)	4.274 (0.561)	6.633 (0.503)
EWK	32.794 (0.498)	9.519 (0.488)	32.674 (0.499)	9.082 (0.488)
EWJ	2.114 (0.444)	2.096 (0.364)	2.103 (0.445)	1.976 (0.364)
EWM	3.654 (0.201)	5.845 (0.312)	2.426 (0.201)	3.172 (0.333)
EWN	4.868 (0.497)	4.936 (0.452)	3.578 (0.498)	4.877 (0.452)
EWY	4.906 (0.474)	9.321 (0.478)	4.068 (0.475)	9.024 (0.479)
EWP	18.505 (0.528)	6.077 (0.473)	18.514 (0.528)	6.179 (0.473)
EWD	6.640 (0.362)	8.023 (0.438)	6.516 (0.362)	8.157 (0.441)
EWL	1.709 (0.481)	2.992 (0.476)	1.580 (0.482)	2.948 (0.476)
EWT	6.758 (0.238)	11.455 (0.366)	6.502 (0.238)	7.903 (0.371)
EWU	2.028 (0.523)	3.762 (0.487)	1.975 (0.524)	4.540 (0.487)
EWV	10.837 (0.320)	17.131 (0.361)	11.004 (0.320)	17.354 (0.361)
FXI	6.925 (0.590)	14.714 (0.549)	4.462 (0.592)	11.965 (0.551)
PGJ	16.579 (0.311)	35.927 (0.244)	15.527 (0.311)	30.558 (0.246)

Note: The out-of-sample forecasts are presented for the 21 sample ETFs from 6 January 2012 to 24 May 2013; the model parameters are estimated using data from 7 January 2000 to 30 December 2011. The entries are mean standard errors (MSE) for the out-of-sample predictions, based upon one-week, and five-week out-of-sample RRV prediction horizons. Figures in parentheses are out-of-sample R-squared (OSR). The full name of 21 ETFs is presented in Table A1 of Appendix A.

contribution in the volatility–volume relation, but also the future researches aiming at volatility modeling and forecasting. Using HAR models to predict volatility from daily data across the 21 single country ETFs, we propose to choose the efficient model for forecasting ETF

volatility. We find that the HAR–RRV–RBV–cum–Vol model is a more efficient method for predicting volatility than the HAR–RRV–RBV model, and such results dominate almost all MSEs, as well as in-sample and out-of-sample forecasting.

Consistent with Darrat et al. (2003) using intraday trading data and Le and Zurbrugg (2010) using EGARCH model for stocks, our results evidence in comport with the sequential information arrival (SIA) hypothesis signifying that Vol contains useful information of future ETF volatility. Also, we offer several simple model specifications where the information content of Vol can be utilized to improve the predictive power of the HAR approach. Our study uses daily data to measure volatility on the 21 country ETFs, offering direct evidence on this issue, which will help less sophisticated investors make better investment decisions on ETF trading. Our out-of-sample analysis confirms the superiority of the HAR–RRV–RBV–cum–Vol model in almost all cases.

Owing to the difficulty in evaluation and the non-availability of high-frequency data, many risk management systems continue to use the daily return-based approaches (e.g., Jacob, 2008). Thus, this study uses daily data. It is important to note that our study does not consider how further improvements could be made by accounting for higher frequency data in improving the performance volatility estimation. It has been shown that high-frequency intraday data can facilitate the formulation of a more accurate inter-daily measurement of ex-post volatility (Andersen and Bollerslev, 1998). For this reason it may be worthwhile as a future research endeavor to investigate how efficient intraday data are for forecasting ETF volatility. Empirical models with high frequency risks remain a useful way for economists and policy makers to clarify the messages in ETFs prices.

Acknowledgment

We would like to thank the Editor and two anonymous referees for their highly constructive comments.

Appendix A

Table 5

Out-of-sample forecasting performance test using the modified DM test.

ETF	Horizons	
	1 week	5 weeks
EWA	7.701***	3.152***
EWK	0.621	0.801
EWO	0.420	–1.187
EWZ	8.965***	3.267***
EWQ	6.661***	9.168***
EWG	0.895	0.480
EWJ	2.753***	–2.955
EWI	2.444***	1.610*
EWK	0.808	1.413*
EWJ	2.724***	1.555*
EWM	19.445***	18.281***
EWN	6.017***	0.524
EWY	11.555***	2.432***
EWP	–0.139	–0.544
EWD	3.922***	–0.538
EWL	6.458***	1.877**
EWT	2.339**	8.707***
EWU	3.442***	–3.707
EWV	–0.927	–0.895
FXI	8.489***	4.122***
PGJ	4.499***	4.595***

Notes: The modified DM test *t*-statistics are presented for the out-of-sample forecasting performance of the 21 sample ETFs covering the period from 6 January 2012 to 24 May 2013, with data from 6 January 2000 to 30 December 2011 being used to estimate the model parameters. The entries are based upon one-week, and five-week out-of-sample RRV prediction horizons. The HAR–RRV–RBV models being taken as the benchmark models for all horizons in the modified DM test. If the test-statistic is positive (negative), this indicates that the model under examination is more (less) accurate than the benchmark model. ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively. The full name of 21 ETFs is presented in Table A1 of Appendix A.

Table A1

Details of the 21 sample ETF data.

No.	Symbol	Full title of ETFs	No. of Obs.
1	EWA	iShare MSCI Australia	3362
2	EWK	iShare MSCI Belgium Capped	3052
3	EWO	iShare MSCI Australia Capped	3200
4	EWZ	iShare MSCI Brazil Capped	3166
5	EWQ	iShare MSCI Canada IDX	3352
6	EWQ	iShare MSCI France	3363
7	EWG	iShare MSCI Germany	3363
8	EWI	iShare MSCI Hong Kong	3363
9	EWI	iShare MSCI Italy Capped	3341
10	EWJ	Capped iShare MSCI Japan ETF	3363
11	EWM	iShare MSCI Malaysia	3362
12	EWN	iShare MSCI Netherlands	3310
13	EWY	iShare MSCI South Korea	3254
14	EWP	iShares MSCI Spain	3328
15	EWD	iShares MSCI Sweden	3315
16	EWL	iShares MSCI Swaziland	3356
17	EWT	iShare MSCI Taiwan	3214
18	EWU	iShare MSCI UK	3363
19	EWV	iShare MSCI Mexico	3362
20	FXI	iShare Xinhua China 25	2171
21	PGJ	Powershares Golden Dragon Halter USX China Portfolio	2127

Notes: Table reports the ticker, name, exchange market, and inception date of the 20 iShares Morgan Stanley Capital International (MSCI) country funds and 3 iShares Broad US equity market funds.

References

- Aber, J.W., Li, D., Can, L., 2009. Price volatility and tracking ability of ETFs. *J. Asset Manag.* 10 (4), 210–221.
- Ackert, L.F., Tian, Y.S., 2008. Arbitrage, liquidity, and the valuation of exchange traded funds. *Financ. Markets, Inst. Inst.* 17 (5), 331–362.
- Akay, O., Griffiths, M.D., Winters, D.B., 2010. On the robustness of range-based volatility estimators. *J. Financ. Res.* 33, 179–199.
- Alizadeh, S., Brandt, M.W., Diebold, F.X., 2002. Range-based estimation of stochastic volatility models. *J. Financ.* 57 (3), 1047–1091.
- Andersen, T.G., 1996. Return volatility and trading volume: an information flow interpretation of stochastic volatility. *J. Financ.* 51, 169–204.
- Andersen, T.G., Bollerslev, T., 1998. Answering the skeptics: yes, standard volatility models do provide accurate forecasts. *Int. Econ. Rev.* 39, 885–905.
- Andersen, T.G., Bollerslev, T., Diebold, F.X., Ebens, J., 2001. The distribution of realized stock return volatility. *J. Financ. Econ.* 61, 43–76.
- Andersen, T.G., Bollerslev, T., Diebold, F.X., 2007. Roughing it up: including jump components in the measurement, modeling and forecasting of return volatility. *Rev. Econ. Stat.* 89, 701–720.
- Bannouh, K., Martens, M., van Dijk, D., 2013. Forecasting volatility with the realized range in the presence of noise and non-trading. *N. Am. J. Econ. Finan.* 26, 535–551.
- Barndorff-Nielsen, O.E., Shephard, N., 2004. Power and bi-power variation with stochastic volatility and jumps. *J. Financ. Econ.* 2, 1–37.
- Barndorff-Nielsen, O.E., Shephard, N., 2006. Impact of jumps on returns and realised variances: econometric analysis of time-deformed Lévy processes. *J. Econ.* 131 (1), 217–252.
- Barnhart, S.W., Rosenstein, S., 2010. Exchange-traded fund introductions and closed-end fund discounts and volume. *Financ. Rev.* 45 (4), 973–994.
- Brunnermeier, M.K., 2009. Deciphering the liquidity and credit crunch 2007–2008. *J. Econ. Theory* 119 (1), 128–150.
- Byun, S.J., Kim, J.S., 2013. The information content of risk-neutral skewness for volatility forecasting. *J. Empir. Financ.* 23, 142–161.
- Çelik, S., Ergin, H., 2014. Volatility forecasting using high frequency data: evidence from stock markets. *Econ. Model.* 36, 176–190.
- Chaboud, A.P., Chiouine, B., Hjalmarsson, E., Loretan, M., 2010. Frequency of observation and the estimation of integrated volatility in deep and liquid financial markets. *J. Empir. Financ.* 17 (2), 212–240.
- Chan, K.C., Fung, H.G., Leung, W.K., 2004. Daily volatility behavior in Chinese futures markets. *J. Int. Financ. Mark. Inst. Money* 14, 491–505.
- Chelley-Steeley, P., Park, K., 2011. Intraday patterns in London listed Exchange Traded Funds. *Int. Rev. Financ. Anal.* 20 (5), 244–251.
- Christensen, K., Podolskij, M., 2006. Range-based estimation of quadratic variation. Working Paper. Aarhus School of Business.
- Chung, S.L., Tsai, W.C., Wang, Y.H., Weng, P.S., 2011. The information content of the S&P 500 index and VIX options on the dynamics of the S&P 500 index. *J. Futur. Mark.* 31 (12), 1170–1201.
- Clark, P.K., 1973. A subordinated stochastic process model with finite variance for speculative prices. *Econometrica* 41, 135–155.
- Copeland, T., 1976. A model of asset trading under the assumption of sequential information arrival. *J. Financ.* 31, 1149–1168.
- Corsi, F., 2004. A simple long memory model of realized volatility. Manuscript. University of Southern Switzerland.
- Darrat, A.F., Rahman, S., Zhong, M., 2003. Intraday trading volume and return volatility of the DJIA stocks: a note. *J. Bank. Financ.* 27, 2035–2043.
- Darrat, A.F., Zhong, M., Cheng, L.T., 2007. Intraday volume and volatility relations with and without public news. *J. Bank. Financ.* 31 (9), 2711–2729.
- Deville, L., 2008. Exchange traded funds: history, trading and research. *Handbook of Financial Engineering*. Springer, US, pp. 67–98.
- Diebold, F.X., Mariano, R.S., 1995. Comparing Predictive Accuracy. *J. Bus. Econ. Stat.* 13, 253–263.
- Forsberg, L., Ghysels, E., 2007. Why do absolute returns predict volatility so well? *J. Financ. Econ.* 5, 31–67.
- Garman, M.B., Klass, M.J., 1980. On the estimation of security price volatilities from historical data. *J. Bus.* 67–78.
- Ghysels, E., Santa-Clara, P., Valkanov, R., 2005. There is a risk-return trade-off after all. *J. Financ. Econ.* 76 (3), 509–548.
- Hansen, P.R., Lunde, A., 2006. Realized variance and market microstructure noise. *J. Bus. Econ. Stat.* 24, 127–161.
- Harvey, D., Leybourne, S., Newbold, P., 1997. Testing the equality of prediction mean squared errors. *Int. J. Forecast.* 13, 281–291.
- Jacob, J., 2008. Estimation and forecasting of stock volatility with range-based estimators. *J. Futur. Mark.* 28 (6), 561–581.
- Jennings, R., Starks, L., Fellingham, J., 1981. An equilibrium model of asset trading with sequential information arrival. *J. Financ.* 36, 143–161.
- Kosev, M., Williams, T., 2011. Exchange-traded funds. *RBA Bulletin* pp. 51–59 (March).
- Krause, T., Tse, Y., 2013. Volatility and return spillovers in Canadian and U.S. industry ETFs. *Int. Rev. Econ. Financ.* 25, 244–259.
- Lamoureux, C., Lastrapes, W., 1990. Heteroskedasticity in stock return data: volume versus GARCH effects. *J. Financ.* 45, 221–229.
- Le, V., Zurbrugg, R., 2010. The role of trading volume in volatility forecasting. *J. Int. Financ. Mark. Inst. Money* 20, 533–555.
- Lee, B.S., Rui, O., 2002. The dynamic relationship between stock returns and trading volume: domestic and cross-country evidence. *J. Bank. Financ.* 26, 51–78.
- Levy, A., Lieberman, O., 2013. Overreaction of country ETFs to US market returns: intraday vs. daily horizons and the role of synchronized trading. *J. Bank. Financ.* 37, 1412–1421.
- Li, H., Cao, S.N., Wang, Y., 2010. The properties and mechanism of long-term memory in nonparametric volatility. *Physica A* 389 (16), 3254–3259.
- Liu, L., Wan, J., 2012. A study of Shanghai fuel oil futures price volatility based on high frequency data: long-range dependence, modeling and forecasting. *Econ. Model.* 29 (6), 2245–2253.
- Liu, H.C., Chiang, S.M., Cheng, N.Y.P., 2012. Forecasting the volatility of S&P depository receipts using GARCH-type models under intraday range-based and return-based proxy measures. *Int. Rev. Econ. Financ.* 22, 78–91.
- Marshall, B.R., Nguyen, N.H., Visaltanachoti, N., 2013. ETF arbitrage: intraday evidence. *J. Bank. Financ.* 37, 3486–3498.
- Martens, M., van Dijk, D., 2007. Measuring volatility with the realized range. *J. Econ.* 138, 181–207.
- McAleer, M., Medeiros, M.C., 2008. Realized volatility: a review. *Econ. Rev.* 27, 10–45.
- Meric, G., Ratner, M., Meric, I., 2008. Portfolio diversification with country index funds. *Glob. Bus. Financ. Rev.* 13, 1–9.
- Miffre, J., 2007. Country-specific ETFs: an efficient approach to global asset allocation. *J. Asset Manag.* 8 (2), 112–122.
- Mizrach, B., 2008. Jump and cojump risk in subprime home equity derivatives. Working Paper (available at SSRN 1089274).
- Narayan, P.K., Sharma, S.S., 2014. Firm return volatility and economic gains: the role of oil prices. *Econ. Model.* 38, 142–151.
- Narayan, P.K., Narayan, S., Prabheesh, K.P., 2014. Stock returns, mutual fund flows and spillover shocks. *Pac. Basin Financ. J.* 29, 146–162.
- Parkinson, M., 1980. The extreme value method for estimating the variance of the rate of return. *J. Bus.* 53, 61–65.
- Patton, A., 2006. Volatility forecast comparison using imperfect volatility proxies. Working Paper 175. Quantitative Finance Research Centre, University of Technology Sydney.
- Rannou, Y., Barneto, P., 2014. Futures trading with information asymmetry and OTC predominance: another look at the volume/volatility relations in the European Carbon Markets. *Energy Econ.* (forthcoming).
- Rompotis, G.G., 2009. Interfamily competition on index tracking: the case of the vanguard ETFs and index funds. *J. Asset Manag.* 10 (4), 263–278.
- Rompotis, G.G., 2010. Does premium impact Exchange-Traded Funds' returns & quest: evidence from iShares. *J. Asset Manag.* 11 (4), 298–308.
- Shin, S., Soydemir, G., 2010. Exchange-traded funds, persistence in tracking errors and information dissemination. *J. Multinat. Financ. Manag.* 20, 214–234.
- Small, K., Wansley, J., Hood, M., 2012. The impact of security concentration on adverse selection costs and liquidity: an examination of exchange traded funds. *J. Econ. Financ.* 36 (2), 261–281.
- Smirlock, M., Starks, L., 1988. An empirical analysis of the stock price–volume relationship. *J. Bank. Financ.* 12, 31–41.
- Stock, J.H., Watson, M.W., 2004. Combination forecasts of output growth in a seven-country data set. *J. Forecast.* 23, 405–430.
- Todorova, N., Souček, M., 2014. The impact of trading volume, number of trades and overnight returns on forecasting the daily realized range. *Econ. Model.* 36, 332–340.
- Tse, Y., Martinez, V., 2007. Price discovery and informational efficiency of international iShares funds. *Glob. Financ. J.* 18 (1), 1–15.
- Tseng, C.T., Huimin, C., Haung, C.S., 2009. Modeling jump and continuous components in the volatility of oil futures. *Stud. Nonlinear Dyn. Econom.* 13 (3), 1–30.
- Wang, J., Yang, M., 2009. Asymmetric volatility in the foreign exchange markets. *J. Int. Financ. Mark. Inst. Money* 19 (4), 597–615.
- Wang, J., Yang, M., 2013. On the risk return relationship. *J. Empir. Financ.* 21, 132–141.
- Zhang, G., 2007. A model of price, volume and sequential information. *Int. J. Bus. Econ.* 6, 207–223.